

Shuya (Joshua) Liu

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Executive Summary

Robotics research engineer with over 10 years of experience designing, developing, and integrating robotic systems across medical robots, industrial robots, and autonomous systems. Expertise includes ROS2, sensor fusion, real-time control, computer vision, teleoperations, haptics, and AR/VR, with a proven track record of leading prototype development from concept through deployment, translating research into end-to-end robotic systems through cross-disciplinary collaboration and integration.

Experience

Institute of Integrative & Innovative Research, UARK

Robotics Research Engineer

Fayetteville, AR (remote)

Oct 2024 - Present

- Developed a robot teleoperation system for ultrasound scanning with real-time remote control, safety constraints, and haptic feedback for improving and monitoring maternal health.
- Architected robotics software framework for robot-assisted orthopedic surgery spanning instrument tracking, navigation, sensing, and real-time control.
- Prototyped an actuation unit for a continuum dexterous manipulator with Fiber Bragg Gratings for high-precision shape estimation for minimally invasive surgeries.
- Built a stereo-vision baseball-catching robot for the institute's open house demonstration.

Hopkins Extreme Materials Institute, JHU

Research Engineer

Baltimore, MD

Aug 2022 - Dec 2023

- Automated conveyor and UR robots using Siemens PLC.
- Automated laser shock and engraving subsystems and supported X-ray subsystem integration and simulation.

Laboratory for Computational Sensing and Robotics, JHU

Research Engineer

Baltimore, MD

Aug 2022 - Oct 2024

- Integrated a continuum dexterous manipulator with a Universal Robot platform.
- Migrated existing system components to a ROS2 framework.

Laboratory of Biomechanical and Image Guided Surgical Systems, JHU

Postdoctoral Research Fellow

Baltimore, MD

Oct 2021 - Aug 2022

- Developed robotic applications for spinal surgery.
- Designed a capacitive shape sensor for a continuum dexterous manipulator.
- Assisted with 3D scanning applications for total-body photography in skin lesion detection.

Laboratory of Biomechanical and Image Guided Surgical Systems, JHU

Graduate Research Assistant

Baltimore, MD

Jan 2016 - Aug 2021

- Designed reattachable fiducial markers for intraoperative patient-to-CT registration using 3D scanners.
- Developed a cutting toolpath algorithm for cranial implant modification and implemented robotic 3D scanning on a KUKA iiwa robot.
- Designed a 5-axis laser cutting system, including mechanical design, GUI development, ROS2 nodes, implant registration, and toolpath optimization.
- Developed a miniature optical tracking system using a RealSense camera for projection mapping and patient-to-CT registration.

BMW Group

Robotics Summer Intern - Internship

Shanghai

May 2014 - Aug 2014

- Contributed to ROS framework development for a self-driving car research platform.
- Developed an RViz visualization plugin for laser radar sensors.

Education

Johns Hopkins University

Baltimore, MD

Ph.D. - Mechanical Engineering

Jan 2016 - Aug 2021

Dissertation: *Robotic Implant Modification for Neuroplastic Surgery*

Advisor: Dr. Mehran Armand

Johns Hopkins University

Baltimore, MD

M.S.E. - Mechanical Engineering

Aug 2013 - May 2015

Changsha University of Science and Technology

Changsha, Hunan, China

B.S. - Mechanical Engineering

Sep 2009 - Jun 2013

Skills

Mechanical Design (Solidworks); Robotics; Sensors; Sensor Fusion; Computer Vision; Human Machine Interface; ROS2; Medical Imaging; X-RAY; Ultrasound; C/C++; GUI Development; Python; Real-Time Control; Laser Systems; CNC Machining; 3D Printing; PCB; 3D Scanning; CAD/CAM; System Integration

Languages

English

Chinese